REPORT

1. Case 1: random0.irl, random3.irl, random5.irl, and random6.irl.
    a. random0.irl dns response jumps to an offset(=4 in this case) that points into the (first 12 bytes are occupied by the DNS header) Fixed DNS header while processing the record name of the first answer in the response.
    b. random3.irl dns response returns a packet of size (randomly generated) which is smaller than the size of the DNS header(=12 bytes).
    c. random5.irl dns response jumps to an offset (randomly generated) that points outside the response packet.
    d. random6.irl dns response encounters a jump to an offset that was already visited resulting in an infinite loop while resolving the CNAME domain name for the first answer in the response.

```
Lookup : random0.irl
Query  : random0.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 1 ms with 82 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 2, authority 0, additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
       random0.irl type 1 class 1
  ------------ [answers] ----------
       random.irl
       ++ invalid record: jump into fixed DNS header
```

```
Lookup : random3.irl
Query  : random3.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 0 ms with 8 bytes
  ++ invalid reply: packet smaller than fixed DNS header
```

```
Lookup : random5.irl
Query  : random5.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 1 ms with 71 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 2, authority 0, additional 0
  succeeded with Rcode = 0
  ------------ [questions] ----------
       random5.irl type 1 class 1
  ------------ [answers] ----------
       random.irl A 1.1.1.1 TTL = 30
       ++ invalid record: jump beyond packet boundary
```

```
Lookup : random6.irl
```

```
Query  : random6.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*******************************
Attempt 0 with 29 bytes... response in 0 ms with 59 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 2, authority 0, additional 0
  succeeded with Rcode = 0
  ---------- [questions] ----------
       random6.irl type 1 class 1
  ---------- [answers] ----------
       random6.irl CNAME
       ++ invalid record: jump loop
```

2. Case 2: random1.irl
   random1.irl denotes 65535 additional entries in the header where the packet only contains 11 additional resource records.

```
Lookup : random1.irl
Query  : random1.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*******************************
Attempt 0 with 29 bytes... response in 0 ms with 468 bytes
  TXID 0xFFFF, flags 0x8600, questions 1, answers 1, authority 0, additional 65535
  succeeded with Rcode = 0
  ---------- [questions] ----------
       random1.irl type 1 class 1
  ---------- [answers] ----------
       random.irl A 1.1.1.1 TTL = 30
  ---------- [additional] ----------
       Episode.IV A 2.2.2.2 TTL = 30
       A.NEW.HOPE A 2.2.2.2 TTL = 30
       It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
       Rebel.spaceships A 2.2.2.2 TTL = 30
       striking.from.a.hidden.base A 2.2.2.2 TTL = 30
       have.won.their.first.victory A 2.2.2.2 TTL = 30
       against.the.evil.Galactic.Empire A 2.2.2.2 TTL = 30
       During.the.battle A 2.2.2.2 TTL = 30
       Rebel.spies.managed A 2.2.2.2 TTL = 30
       to.steal.secret.plans A 2.2.2.2 TTL = 30
       to.the.Empires.ultimate.weapon A 2.2.2.2 TTL = 30
       ++ invalid section : not enough records
```

3. Case 3: random7.irl
   random7.irl dns response is truncated at the byte denoting a jump while resolving the domain name for the CNAME of the first answer, a jump signal contains of two bytes with first two bits set denoting the jump and the rest 14 bits representing the offset for the jump, in this case the packet is truncated after the first byte.

```
Lookup : random7.irl
Query  : random7.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 0 ms with 42 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 2, authority 0, additional 0
  succeeded with Rcode = 0
  ---------- [questions] ----------
       random7.irl type 1 class 1
  ---------- [answers] ----------
       random7.irl CNAME
       ++ invalid record: truncated jump offset
```

4. Case 4: random4.irl:
   In random4.irl dns response the server randomly truncates the packet. Three errors that aren't produced in the above responses,
   a. ++ invalid record: truncated name - it occurs when the truncation happens in the resource name of a resource record.
   b. ++ invalid record: truncated RR answer header - it occurs when the response is truncated after the record name and before the answer data starts.
   c. ++ invalid record: truncated RR answer header - it occurs when the truncation happens within 4 bytes after the record header as the entries are of type A answer data is of 4 bytes containing the IP address which gets truncated.

```
Lookup : random4.irl
Query  : random4.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 1 ms with 359 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 1, authority 0, additional 11
  succeeded with Rcode = 0
  ---------- [questions] ----------
       random4.irl type 1 class 1
  ---------- [answers] ----------
       random.irl A 1.1.1.1 TTL = 30
  ---------- [additional] ----------
       Episode.IV A 2.2.2.2 TTL = 30
       A.NEW.HOPE A 2.2.2.2 TTL = 30
       It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
       Rebel.spaceships A 2.2.2.2 TTL = 30
       striking.from.a.hidden.base A 2.2.2.2 TTL = 30
       have.won.their.first.victory A 2.2.2.2 TTL = 30
       against.the.evil.Galactic.Empire A 2.2.2.2 TTL = 30
       During.the.battle A 2.2.2.2 TTL = 30
       Rebel.sp
       ++ invalid record: truncated name
```

```
Lookup : random4.irl
Query  : random4.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 0 ms with 214 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 1, authority 0, additional 11
  succeeded with Rcode = 0
  ------------ [questions] ----------
       random4.irl type 1 class 1
  ------------ [answers] ----------
       random.irl A 1.1.1.1 TTL = 30
  ------------ [additional] ----------
       Episode.IV A 2.2.2.2 TTL = 30
       A.NEW.HOPE A 2.2.2.2 TTL = 30
       It.is.a.period.of.civil.war A 2.2.2.2 TTL = 30
       Rebel.spaceships A 2.2.2.2 TTL = 30
       striking.from.a.hidden.base
       ++ invalid record: truncated RR answer header
```

```
Lookup : random4.irl
Query  : random4.irl, type 1, TXID 0xFFFF
Server : 127.0.0.1
*********************************
Attempt 0 with 29 bytes... response in 0 ms with 105 bytes
  TXID 0xFFFF, flags 0x8400, questions 1, answers 1, authority 0, additional 11
  succeeded with Rcode = 0
  ------------ [questions] ----------
       random4.irl type 1 class 1
  ------------ [answers] ----------
       random.irl A 1.1.1.1 TTL = 30
  ------------ [additional] ----------
       Episode.IV A 2.2.2.2 TTL = 30
       A.NEW.HOPE
       ++ invalid record: RR value length stretches the answer beyond packet
```

5. Extra credit
   The algorithm overwrites a continuous string of multiple occurrences “lol” over the fixed const response packet, it never overwrites the dns header i.e the starting position is after 12 bytes and the no. of occurrences of “ lol”s are randomly chosen >= 10 and <=39 such that the overwriting string is contained within the packet and does not overflow.